

Department of Chemistry and
Department of Physics & Astronomy
University of Waterloo
Waterloo, ON, Canada
giresse.feugmo@gmail.com
August 8, 2026

The Editors
Corrosion Science
Elsevier

Dear Editors,

I am pleased to submit the manuscript “**Identifiability of a point defect model for oxide growth on AISI 347**” for consideration as a research paper in *Corrosion Science*.

The point defect model is the standard mechanistic description of passive film growth, and fitting it with four to six parameters to a handful of thickness measurements is routine practice. Whether those measurements determine the parameters is almost never tested. This paper carries out that test for a representative case, and the answer is that several of them are not determined—with consequences large enough to change a lifetime assessment.

The system is the duplex oxide grown on Nb-stabilized AISI 347 (X6CrNiNb18-10) in simulated boiling-water-reactor hydrothermal water (240 °C, 7 MPa, 0.4 ppm dissolved O₂). I identify a reduced five-parameter model from published depth-profile data at three exposure times using a differentiable, physics-informed inversion, validated throughout against an exact closed-form reference rather than against another numerical solution.

I believe the work suits *Corrosion Science* for four reasons.

- **It is the first identifiability characterization of a PDM fit.** Profile likelihood, a 1000-member bootstrap, a prior-leverage test and a residual-leverage decomposition are built into the inversion rather than added afterward. Of the five parameters of the accepted model, three exposure times leave one undetermined and pin a second only to within an order of magnitude, and 95% of the fit statistic is carried by three chromium points—80% by a single measurement. These tools are mature in adjacent fields but, to my knowledge, have not been brought to the PDM; the diagnosis transfers to any sparsely sampled degradation model.
- **The consequences are quantitative and actionable.** The identified kinetics fit the calibration data as well as an empirical power law but extrapolate very differently: at ten years the power law predicts a 3.5-to-6.3-fold thicker barrier layer than the mechanistic 535 nm (bootstrap 95% interval 447–704 nm), varying 1.19-to-1.56-fold across the scanned operating envelope. A designed exposure beyond roughly 3000 hours would discriminate the two—a concrete experiment this analysis motivates.
- **A first mechanistic description for this alloy, honestly bounded.** Prior characterization of AISI 347 oxide in BWR water stopped at empirical per-element power-law fits with no shared physical parameter set. Here the Cr-rich barrier and Fe-rich outer layers are connected through a single kinetic description. I am explicit that the model structure is not itself Nb-specific, that the neural inversion is validated against an exact classical reference rather than required by the problem, and that the constant-field closure underlying the derived field strength does not survive a self-consistency test. I would rather state these bounds than have them inferred.
- **A diagnosed and remedied failure mode.** I document an inversion failure specific to

sparse experimental data—the data loss satisfied while the governing equation is not—and remedy it with a forward-solve self-consistency check, a cautionary and constructive result for the growing use of physics-informed learning in corrosion modeling.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. I declare no competing interests, and the work involves no human or animal subjects. All code and the digitized data are released with provenance notes, as described in the Data availability statement.

Thank you for considering this submission.

Sincerely,

Conrard Giresse Tetsassi Feugmo